

25/04/2019

PISA PROJECT

Performance report on the multi-channel device



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1 Introduction

The purpose of this performance report is to validate the proper operation of the new multi-channel devices.

For this purpose, various artificial signals, using a function generator, will be produced in order to visually check the correct functioning of the device in the time and frequency domain.

Various tests will be carried out, first with the DHCP application allowing the live measurement, then with the help of the classic application which records on a USB key.

2 Equipment used

The following devices are used to perform the various tests:

- Multi-channel device "E": for all tests, channel 3 was used.
- Short electrodes 30cm long
- Long electrodes 5m long.
- UNIT-T UTG1005A function generator for generating artificial signals.

3 Tests performed

Five different tests were carried out:

- In the first one, the DHCP application for live measurement was used. Various specific signals were injected with the function generator to check a correct shape during the measurements. The correct amplitudes and frequencies were also checked. Short cables were used for this test.

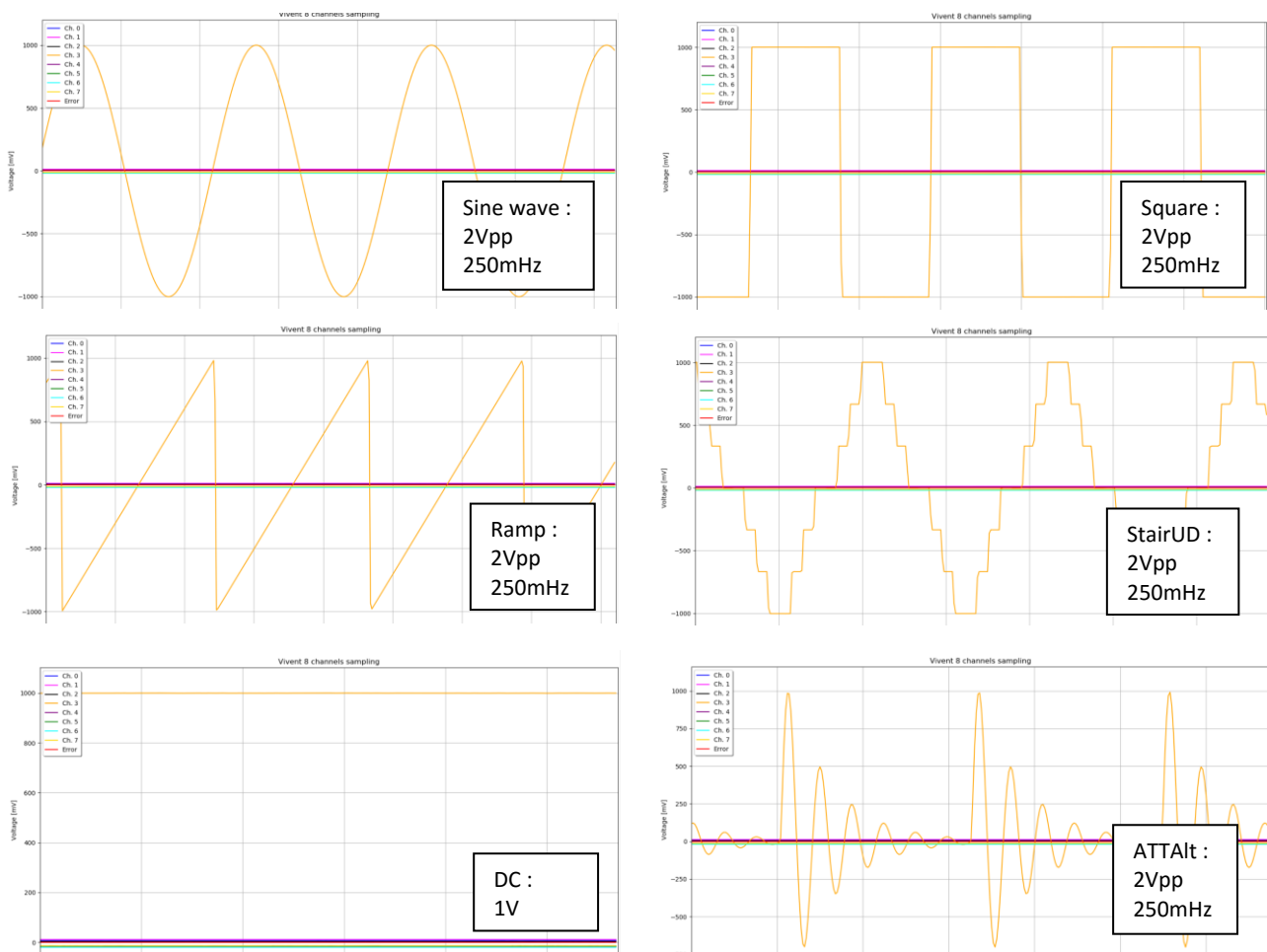
- In the second, the classic application was used to acquire different signals (injected with the function generator) on the USB stick. For each signal, a measurement of approximately 10 minutes was performed. The signals were then observed in the time and frequency domains in order to observe the characteristics of the measurements performed. Short cables were used for this test.
- The third test was exactly the same as the second, but by replacing the short cables with long cables about 5m long.
- The fourth test was simply an in-air measurement, completely removing the cables, to see if these cables influence the number of disturbance peaks.
- The last test was based on the same principle as the third tests, but by connecting 4 other unused channels to plants and pots to determine if there is any cross talk / interference between the cables. Only long cables were used.

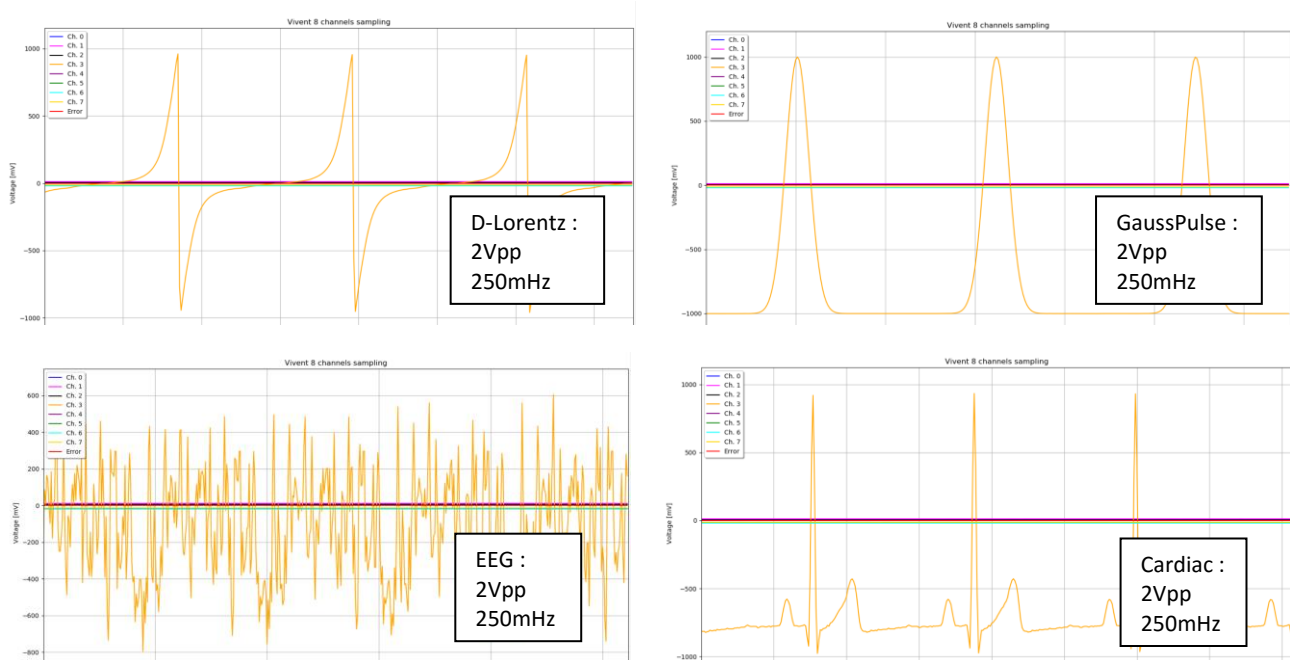
4 Results

The different results obtained are demonstrated in the following sections and separated into 5 chapters.

4.1 DCHP application, live measurement of various specific signals (short cable)

The following captures show the results obtained with the DHCP application for 10 different signals.





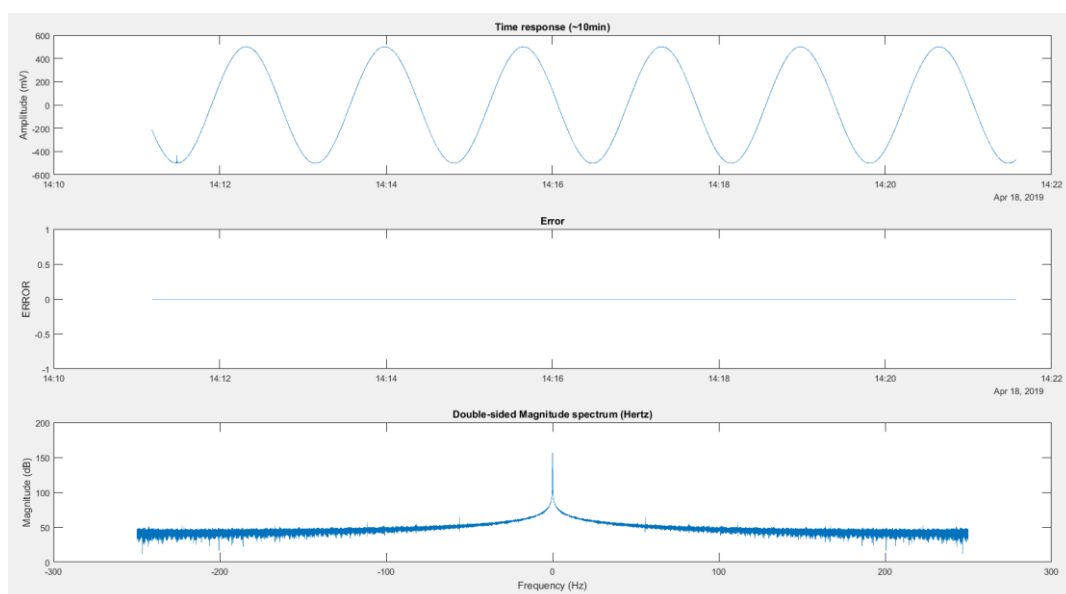
All measured signals have the desired shape. The tests are therefore good and no problems/comments should be reported here.

4.2 Classical application, temporal and frequency analysis of different signals (short cable)

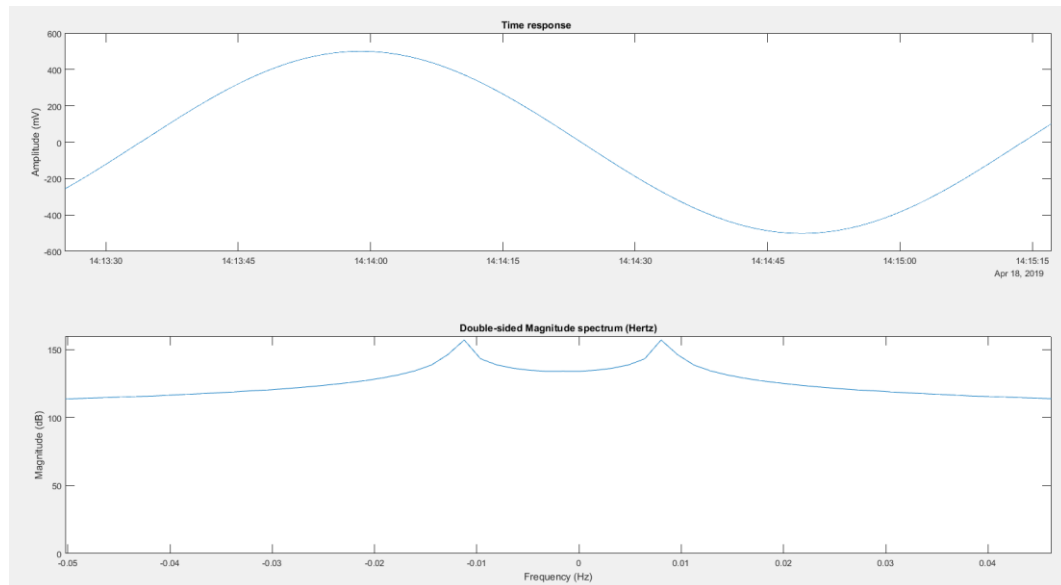
6 different signals have been measured with short cables and analyzed here.

Low frequency sine wave

A sine wave of 0.01Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



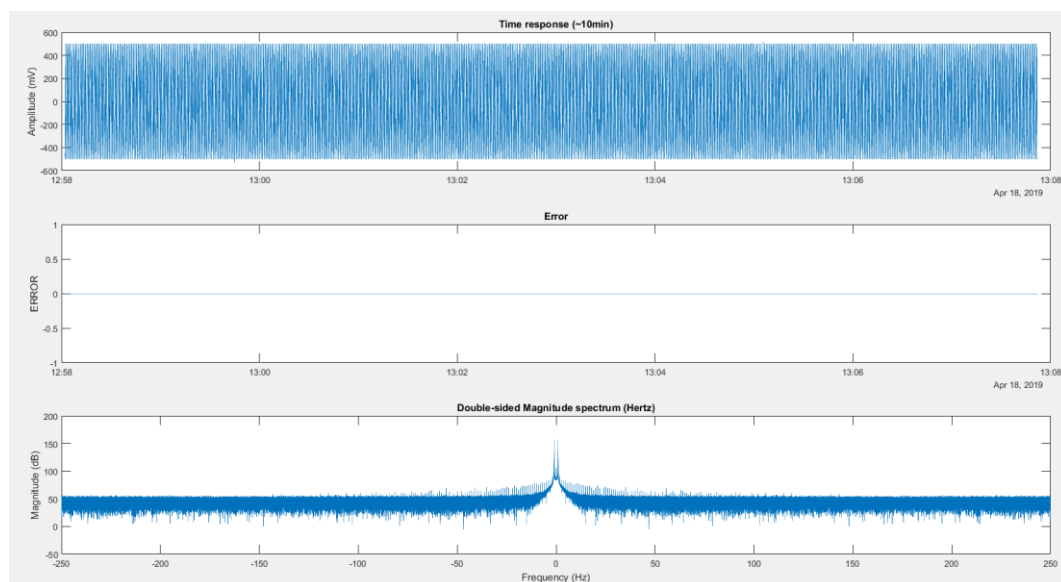
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



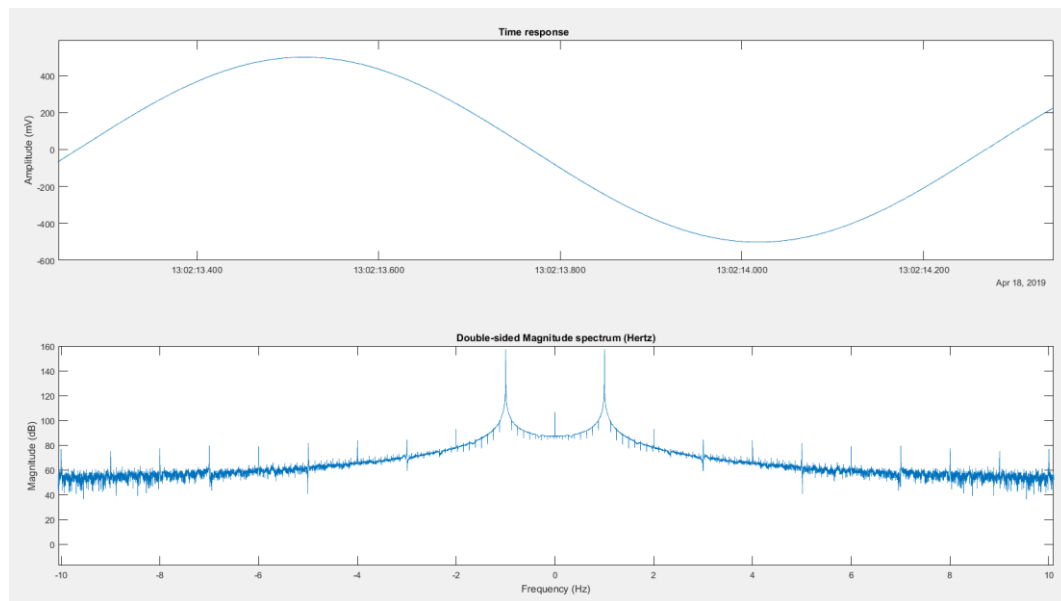
In the time domain, we do observe a period of 100 seconds, therefore a frequency of 0.01Hz. In the frequency domain, we can see the pulses at 0.01Hz which represent the sine wave.

Medium frequency sine wave

A sine wave of 1Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



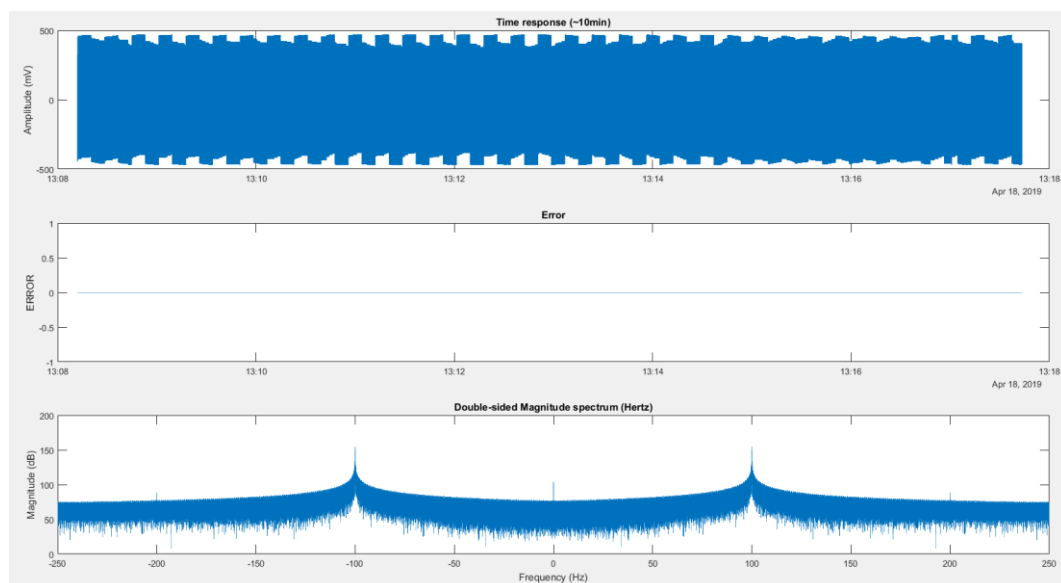
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



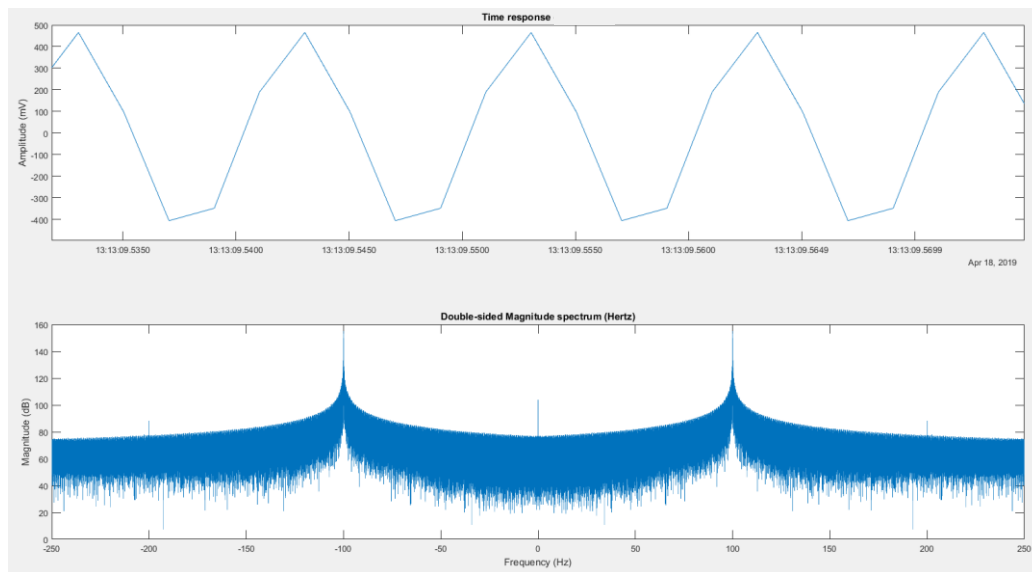
In the time domain, we do observe a period of 1 second, therefore a frequency of 1Hz. In the frequency domain, we can see the pulses at 1Hz which represent the sine wave.

High frequency sine wave

A sine wave of 100Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



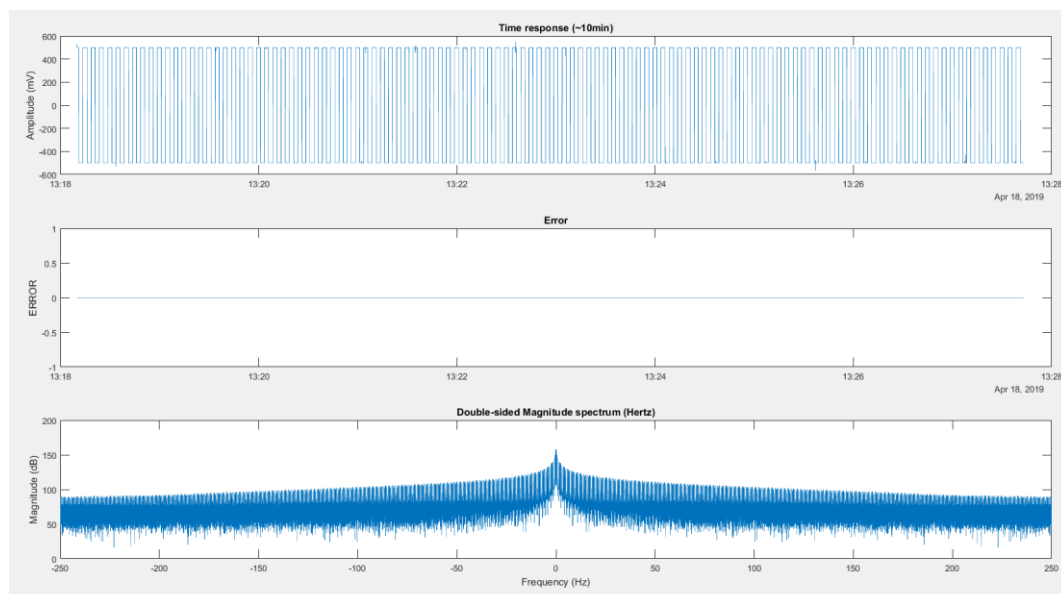
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



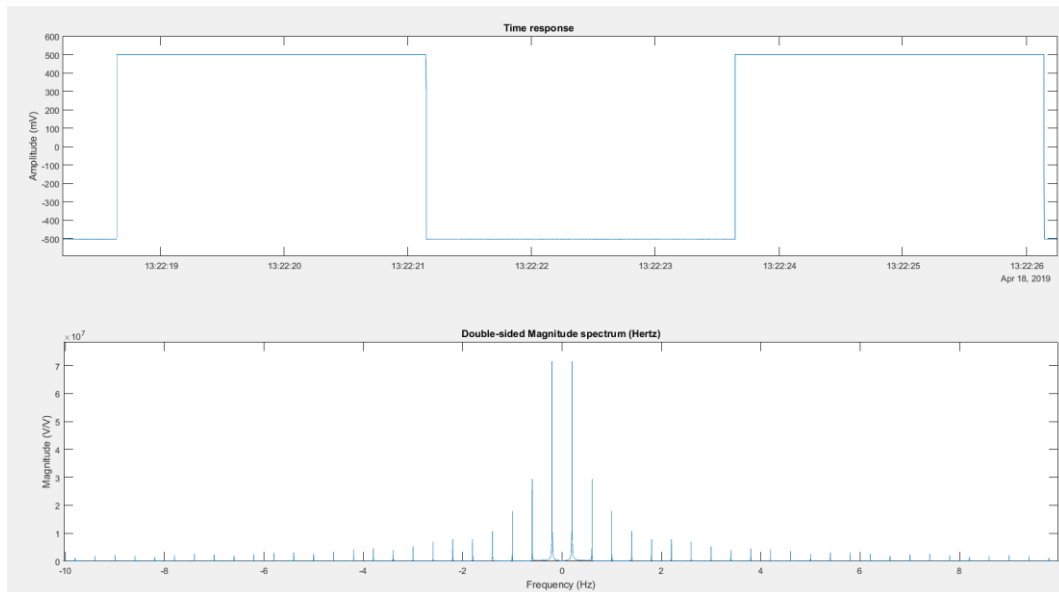
In the time domain, we do observe a period of 0.01 second, therefore a frequency of 100Hz. We observe that the sine wave is no longer really a sine wave. This is due to the fact that a sine at 100Hz is sampled at 500Hz. So we only have 5 samples per period, which is not enough to have a beautiful sinus shape. In the frequency domain, we can see the pulses at 100Hz which represent the sine wave.

Low frequency square wave

A square wave of 0.2Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



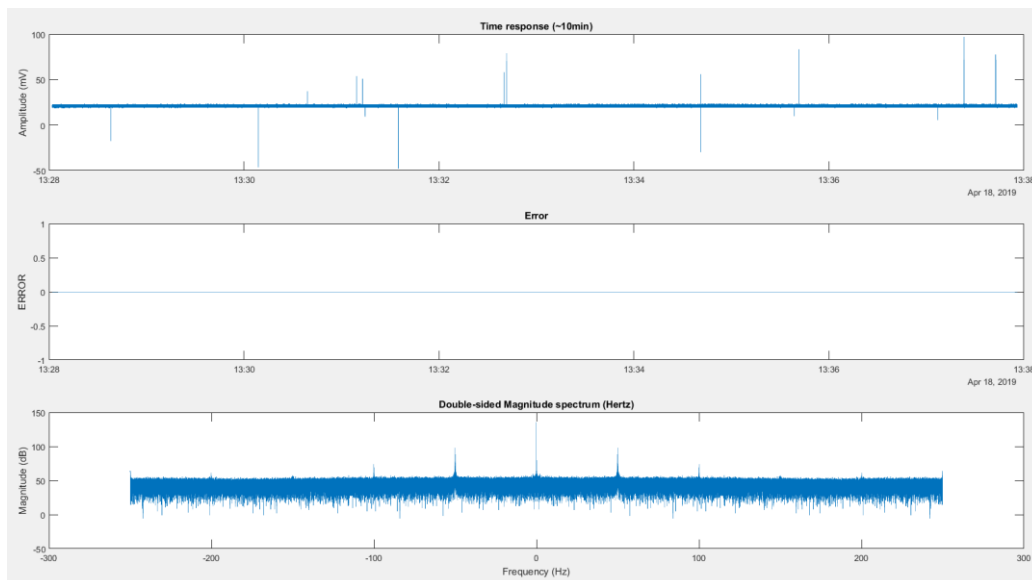
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



In the time domain, we do observe a period of 5 seconds, therefore a frequency of 0.2Hz. In the frequency (here, the result is not in dB !), we can observe that the result is a sinc, which is the fourrier transform of a square wave.

In the air

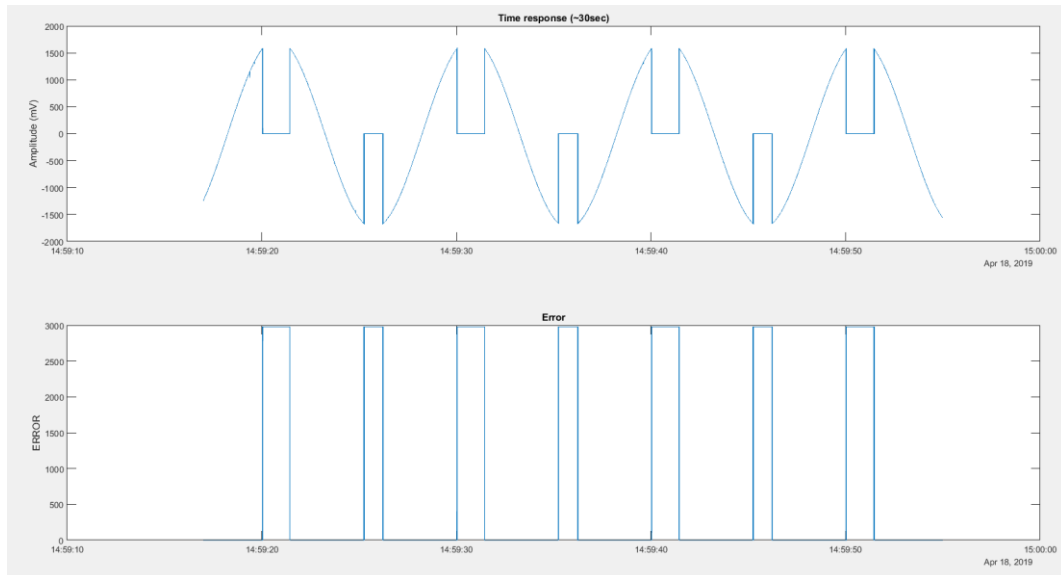
Here, the electrodes were not connected to the function generators and were left in the air to observe what was being measured.



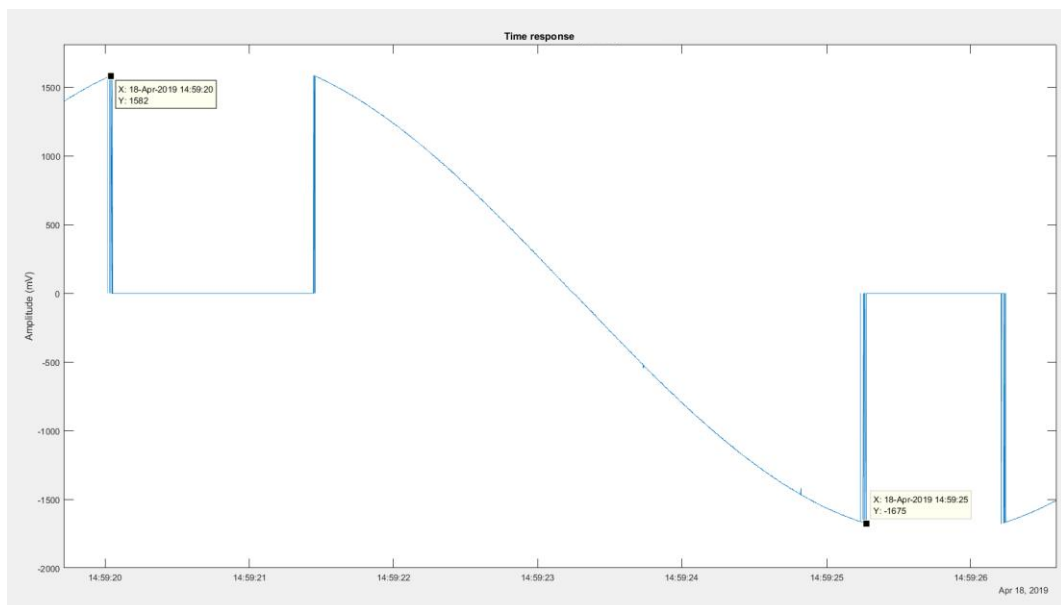
It can be seen that no errors were detected by the device during this period. In the temporal domain, peaks are sometimes observed, certainly due to disturbances on the network and on the cables. In the frequency domain, we can also observe peaks at 50Hz as well as all these harmonics due to our 50Hz network.

Low frequency sine wave with overvoltage

A sinus wave of 0.1Hz and 3.5Vpp was measured here for a duration of 30 seconds. An overvoltage will therefore be caused each time we exceed $\pm 1.65V$. The results in the time domain are shown below:



It can be seen that an error is detected each time the limit voltage is exceeded by $\pm 1.65V$. This has the effect of resulting in a null measurement. A zoom was then performed to see the exact value of this overvoltage :



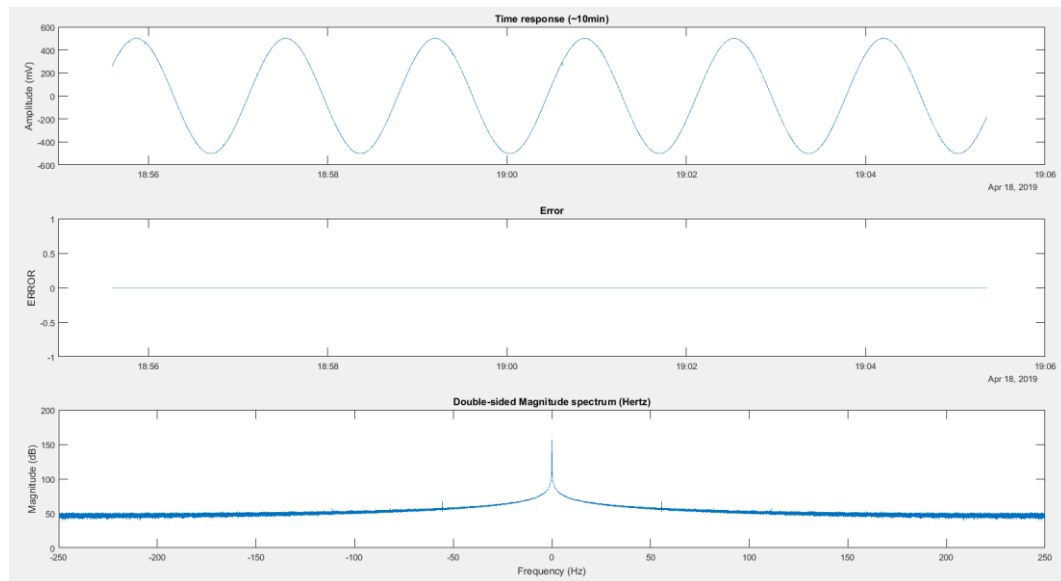
We then notice that the value of the overvoltage is 1.582V and the undervoltage is -1.675V, which does not exactly correspond to the 1.65V specified in the datasheet. This is certainly due to the AD7770's V_{SS} and V_{DD} power supplies, which are not very accurate.

4.3 Classical application, temporal and frequency analysis of different signals (long cable)

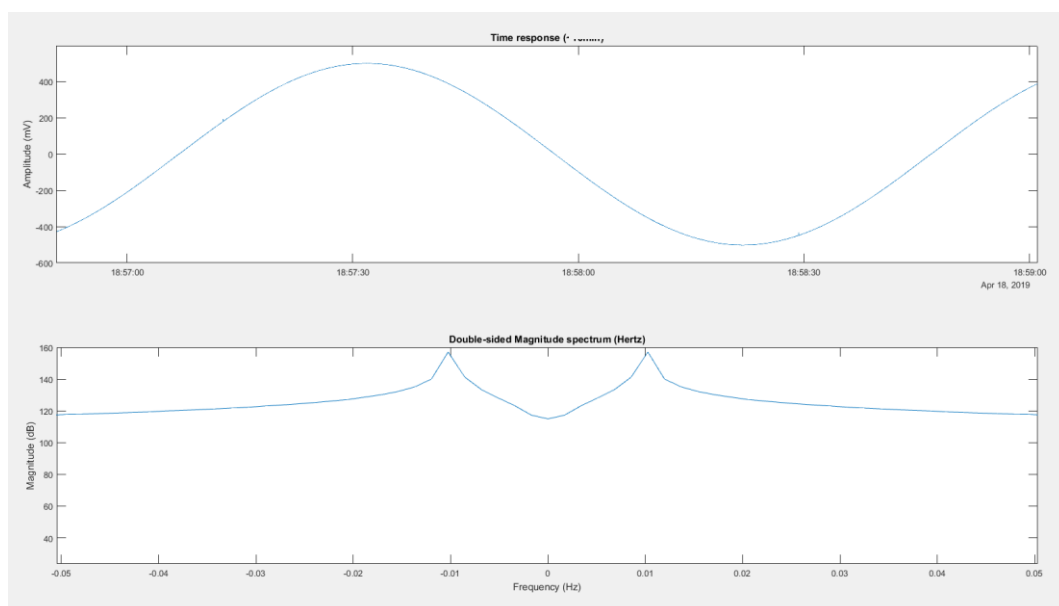
6 different signals have been measured with long cables and analyzed here.

Low frequency sine wave

A sine wave of 0.01Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



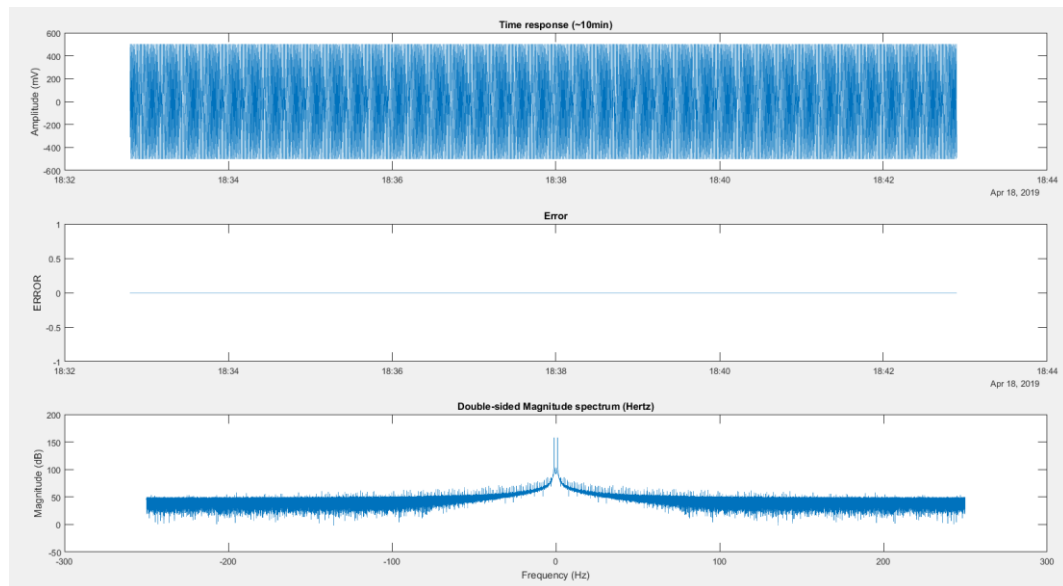
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



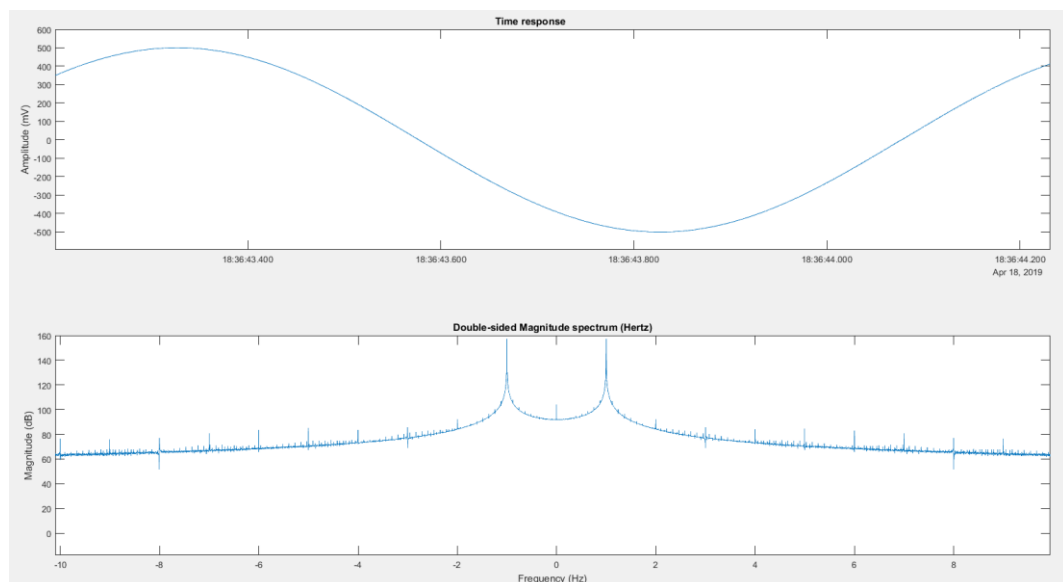
In the time domain, we do observe a period of 100 seconds, therefore a frequency of 0.01Hz. In the frequency domain, we can see the pulses at 0.01Hz which represent the sine wave.

Medium frequency sine wave

A sine wave of 1Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



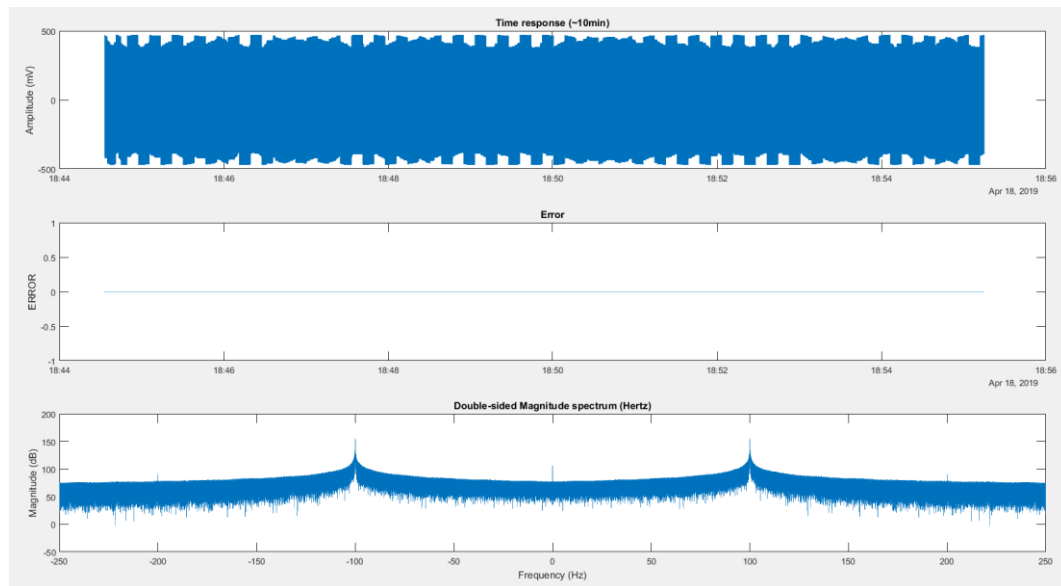
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



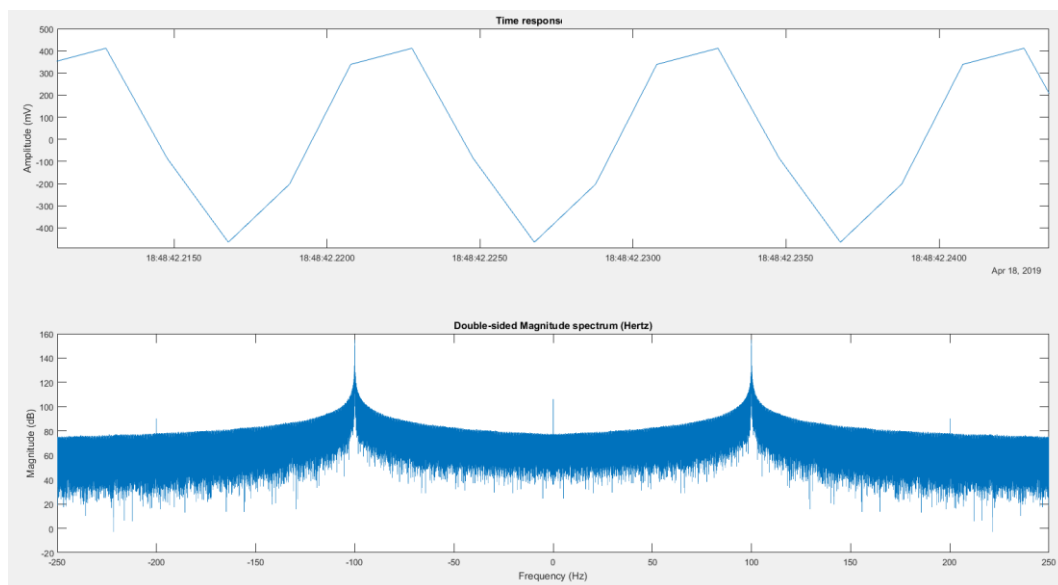
In the time domain, we do observe a period of 1 second, therefore a frequency of 1Hz. In the frequency domain, we can see the pulses at 1Hz which represent the sine wave.

High frequency sine wave

A sine wave of 100Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



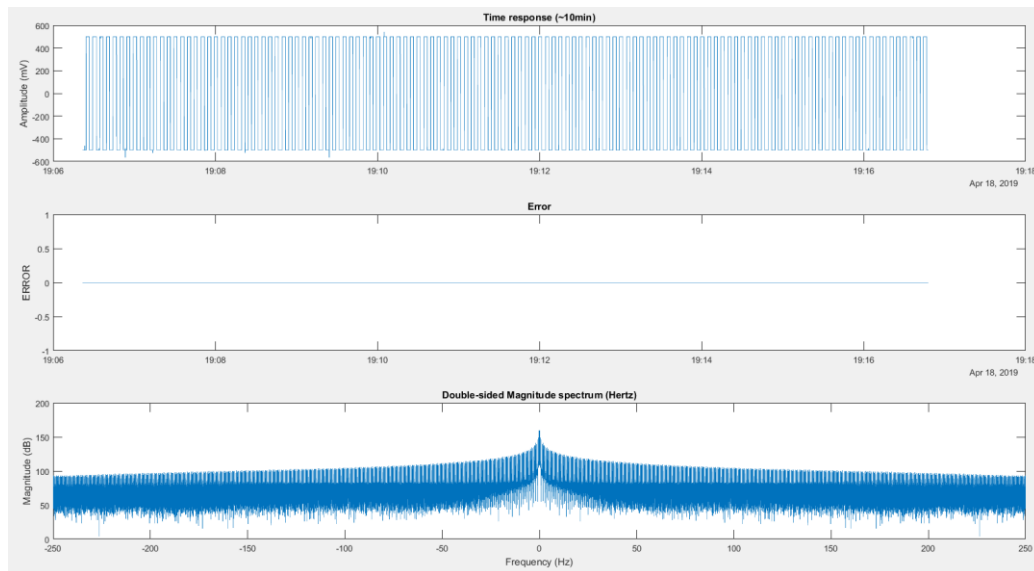
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



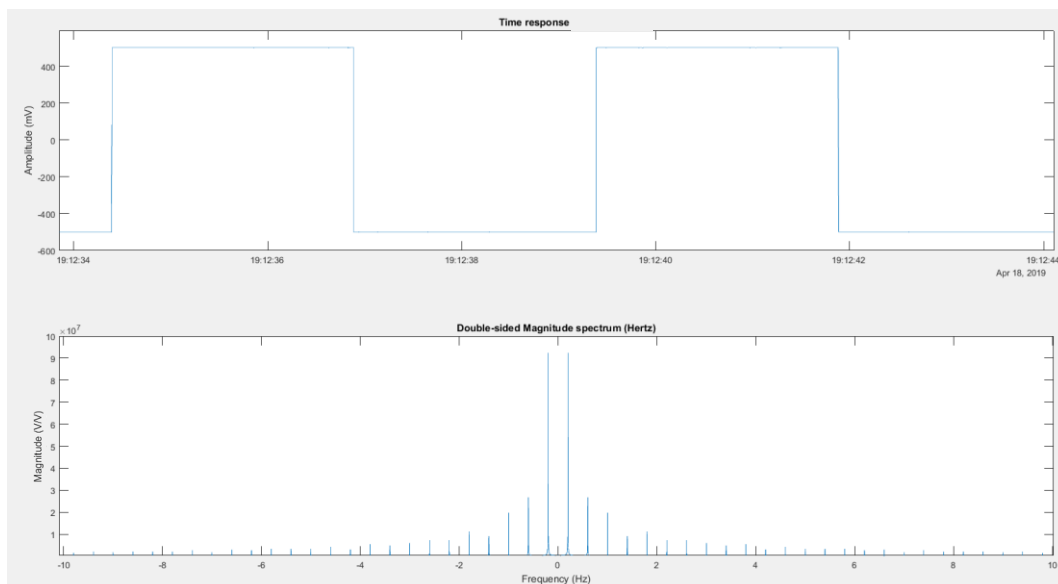
In the time domain, we do observe a period of 0.01 second, therefore a frequency of 100Hz. We observe that the sine wave is no longer really a sine wave. This is due to the fact that a sine at 100Hz is sampled at 500Hz. So we only have 5 samples per period, which is not enough to have a beautiful sinus shape. In the frequency domain, we can see the pulses at 100Hz which represent the sine wave.

Low frequency square wave

A square wave of 0.2Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



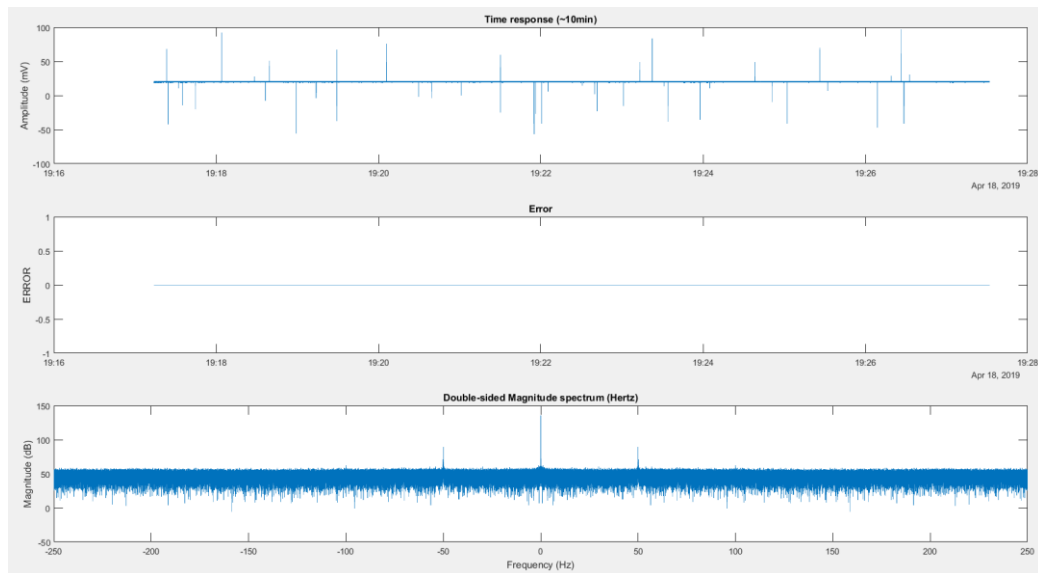
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



In the time domain, we do observe a period of 5 seconds, therefore a frequency of 0.2Hz. In the frequency (here, the result is not in dB !), we can observe that the result is a sinc, which is the fourrier transform of a square wave.

In the air

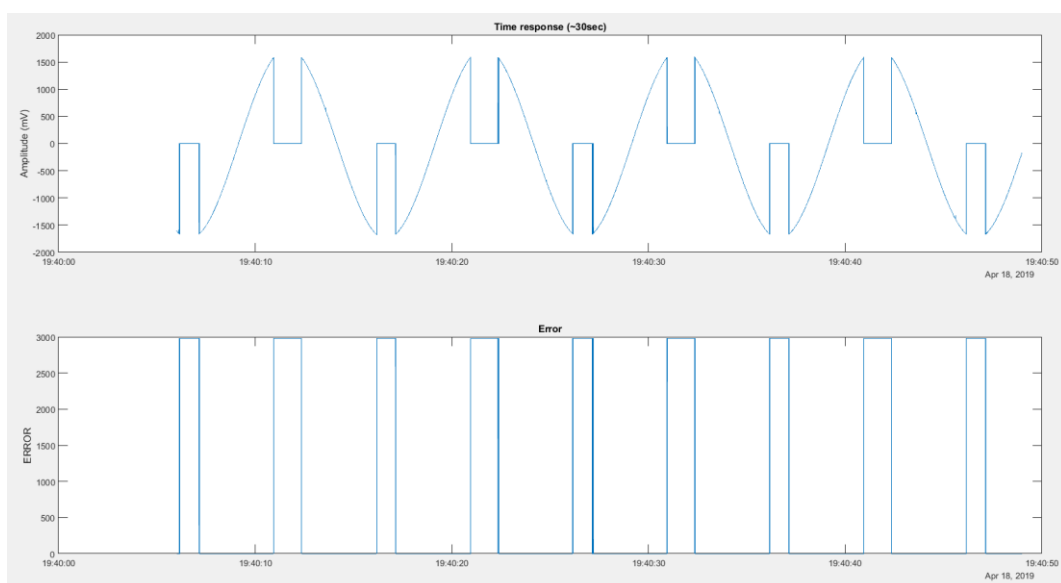
Here, the electrodes were not connected to the function generators and were left in the air to observe what was being measured.



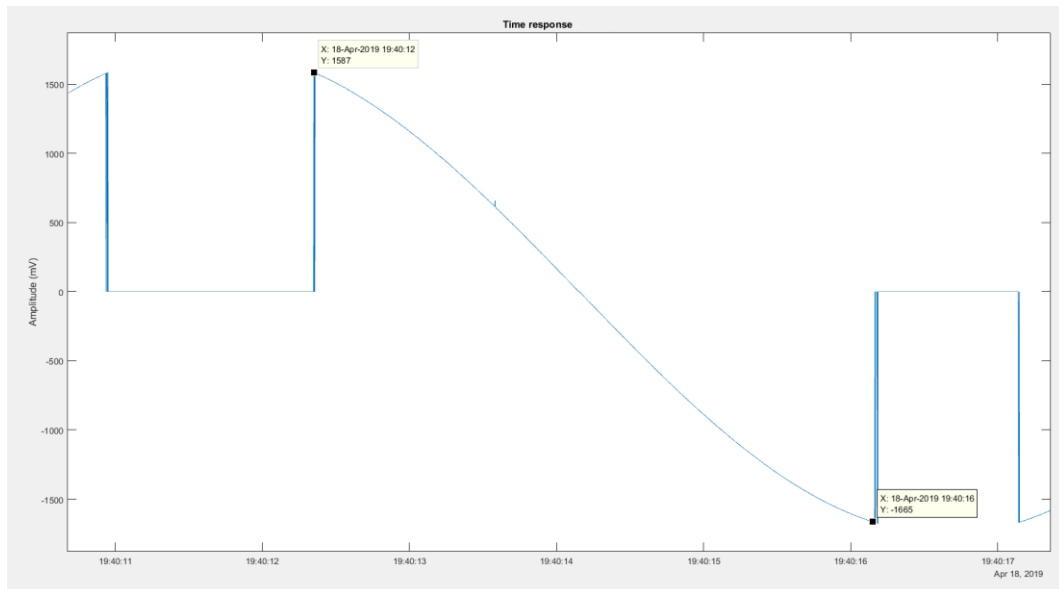
It can be seen that no errors were detected by the device during this period. In the temporal domain, peaks are sometimes observed, certainly due to disturbances on the network and on the cables. Compared to the results in section 4.2, it can be seen that the number of perturbations is significantly higher. In the frequency domain, we can also observe peaks at 50Hz as well as all these harmonics due to our 50Hz network.

Low frequency sine wave with overvoltage

A sinus wave of 0.1Hz and 3.5Vpp was measured here for a duration of 30 seconds. An overvoltage will therefore be caused each time we exceed $\pm 1.65V$. The results in the time domain are shown below:



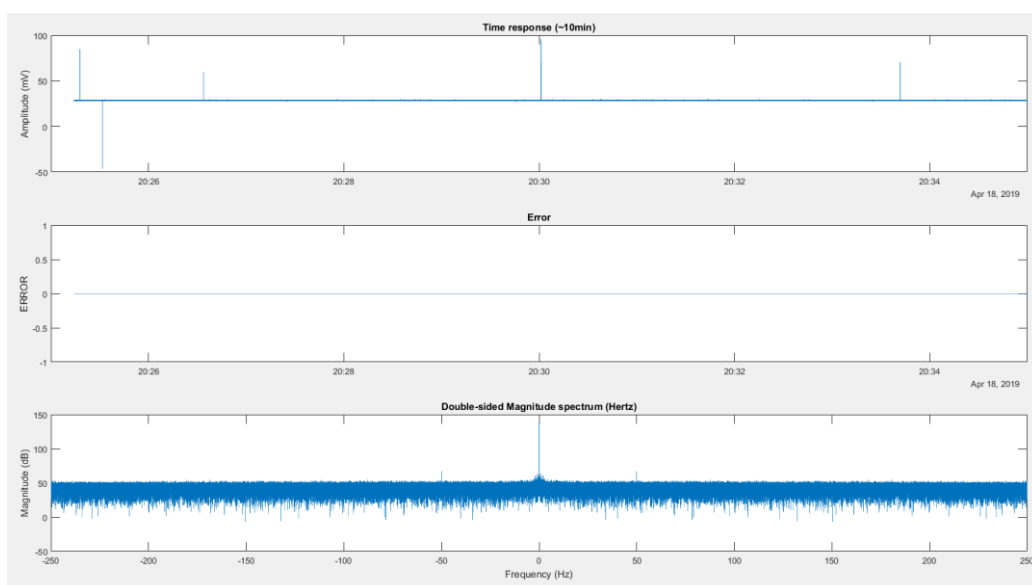
It can be seen that an error is detected each time the limit voltage is exceeded by $\pm 1.65\text{V}$. This has the effect of resulting in a null measurement. A zoom was then performed to see the exact value of this overvoltage :



We then notice that the value of the overvoltage is 1.587V and the undervoltage is -1.665V, which does not exactly correspond to the 1.65V specified in the datasheet. This is certainly due to the AD7770's V_{SS} and V_{DD} power supplies, which are not very accurate.

4.4 Classical application, temporal and frequency analysis of signal "in the air" (no cable)

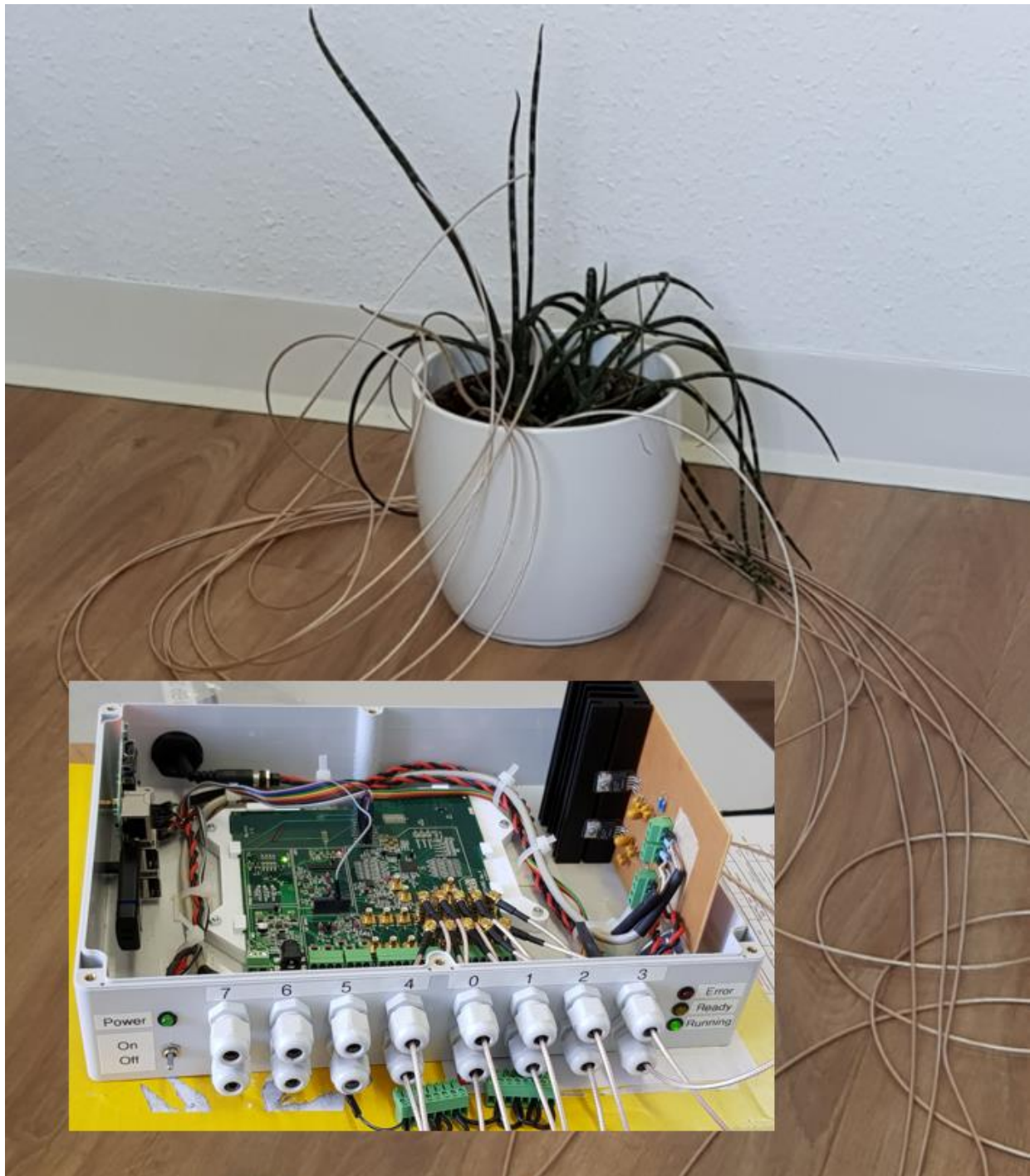
In order to observe whether the cables and their length disturb the signal, a measurement without the cables, with simply measurements in the air, was performed.



It can be seen that the number of perturbations has decreased significantly. It can therefore be concluded that the cables add disturbances.

4.5 Classical application, temporal and frequency analysis of different signals (4 other cables connected)

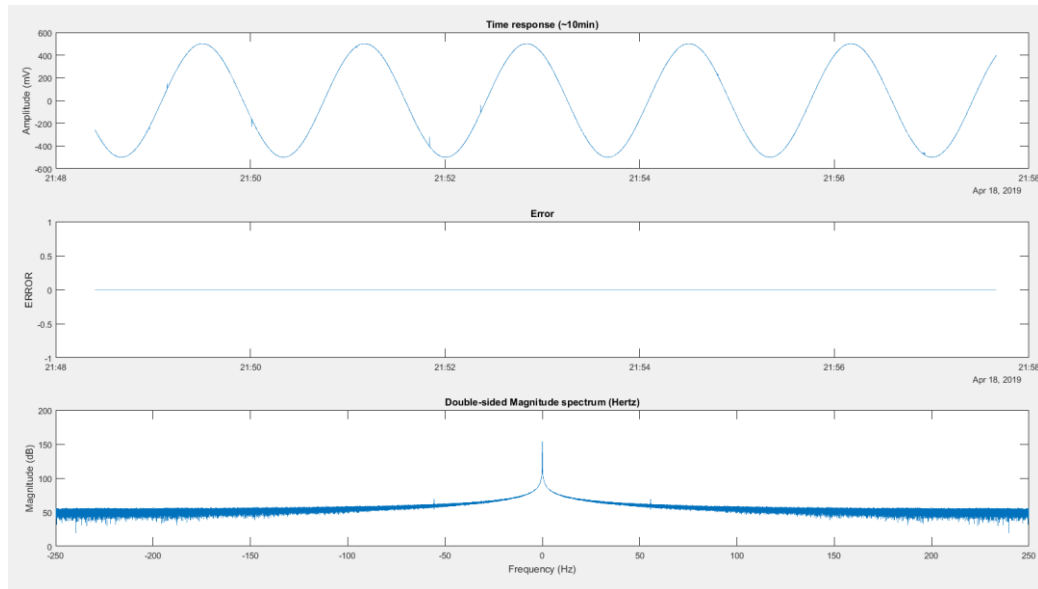
The tests were repeated one last time. Channel 3 is always the one used and connected to the function generator. The 4 neighbouring channels were connected to a plant. The 4 grounds were connected to the pot and the 4 active electrodes were connected to different places on the plant. The following figure illustrates the test bench.



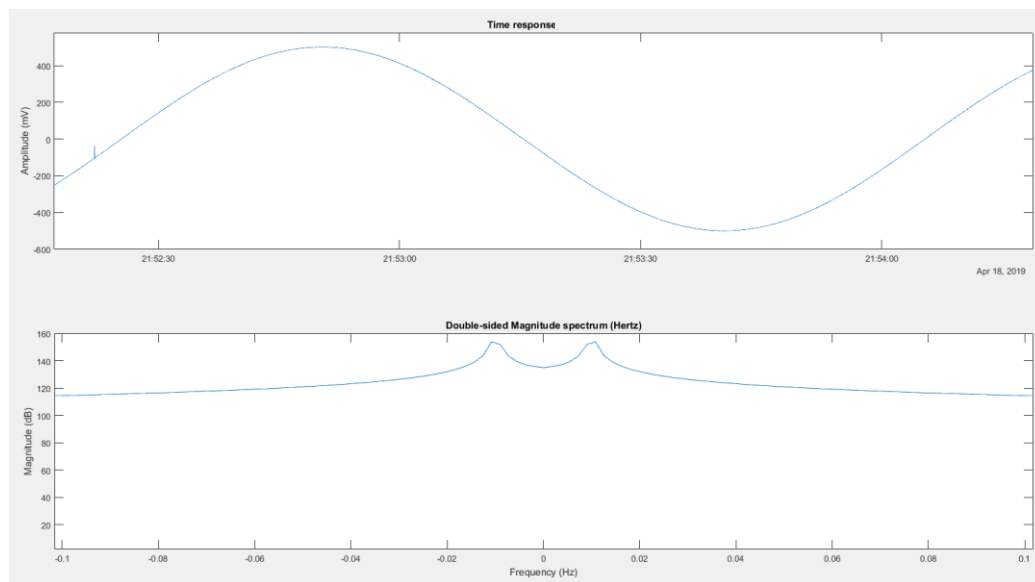
6 different signals have been measured with long cables and analyzed here.

Low frequency sine wave

A sine wave of 0.01Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



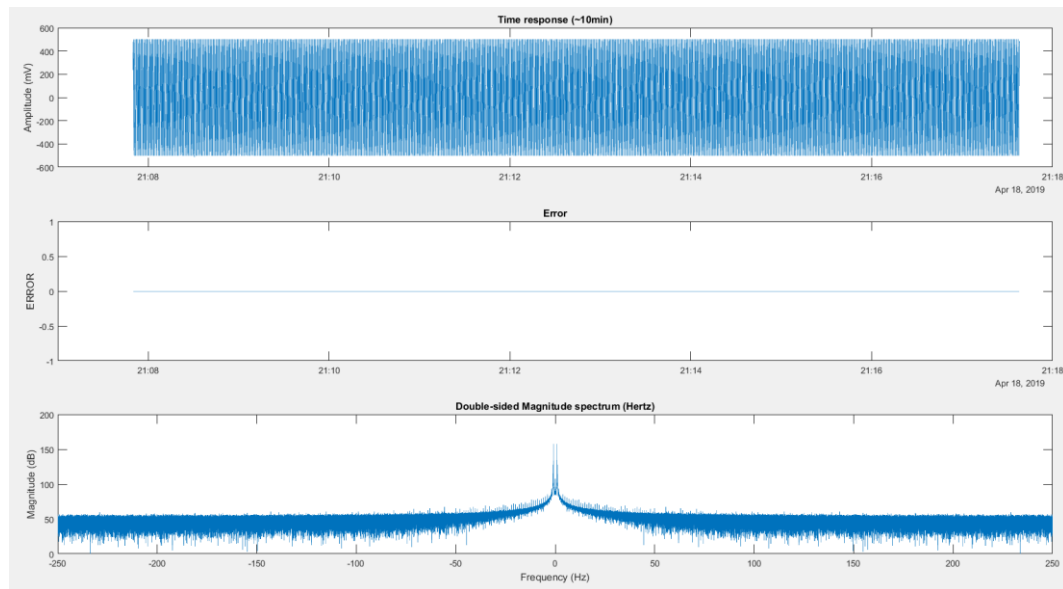
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



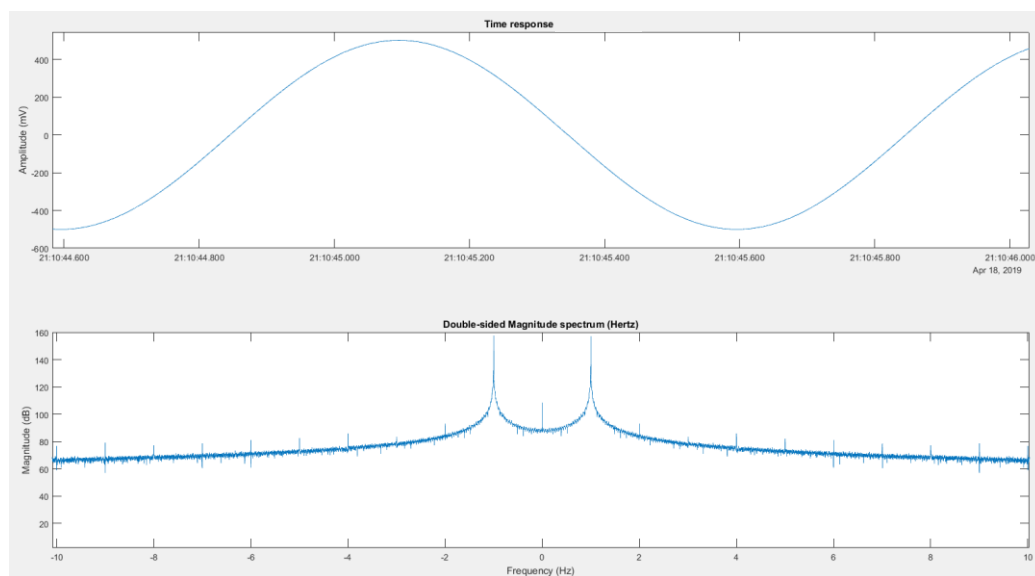
In the time domain, we do observe a period of 100 seconds, therefore a frequency of 0.01Hz. In the frequency domain, we can see the pulses at 0.01Hz which represent the sine wave.

Medium frequency sine wave

A sine wave of 1Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



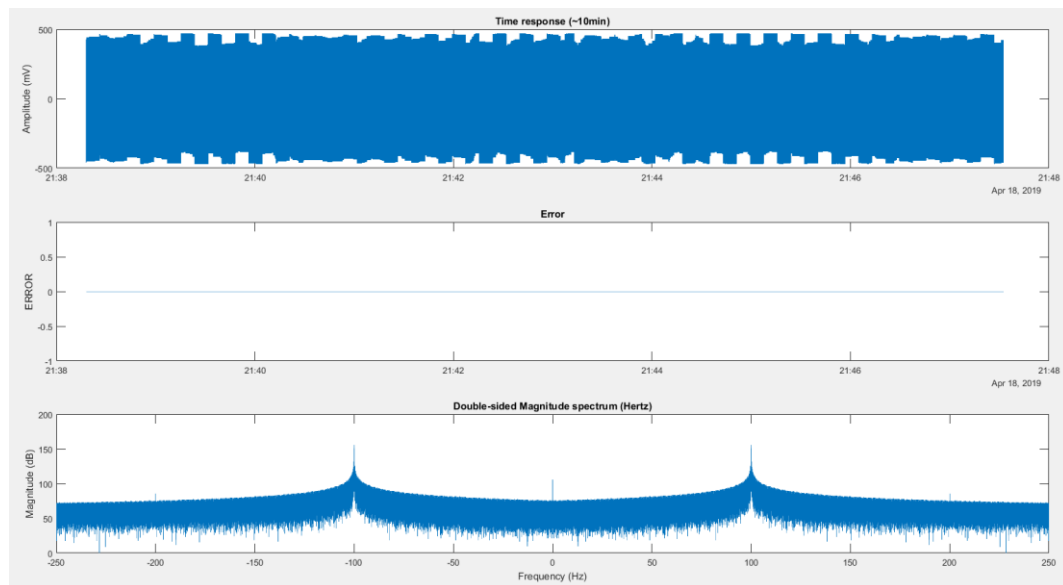
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



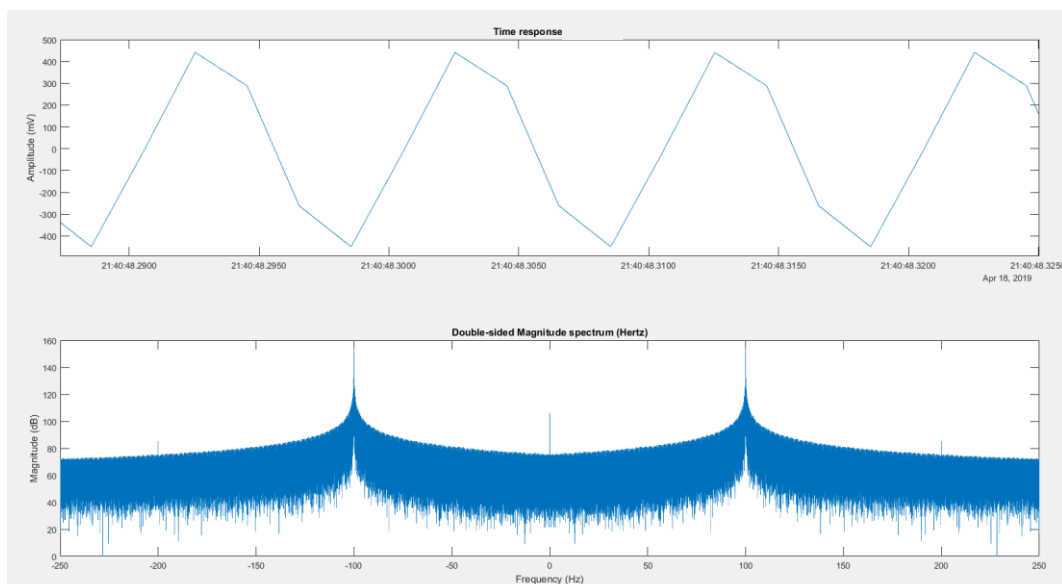
In the time domain, we do observe a period of 1 second, therefore a frequency of 1Hz. In the frequency domain, we can see the pulses at 1Hz which represent the sine wave.

High frequency sine wave

A sine wave of 100Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



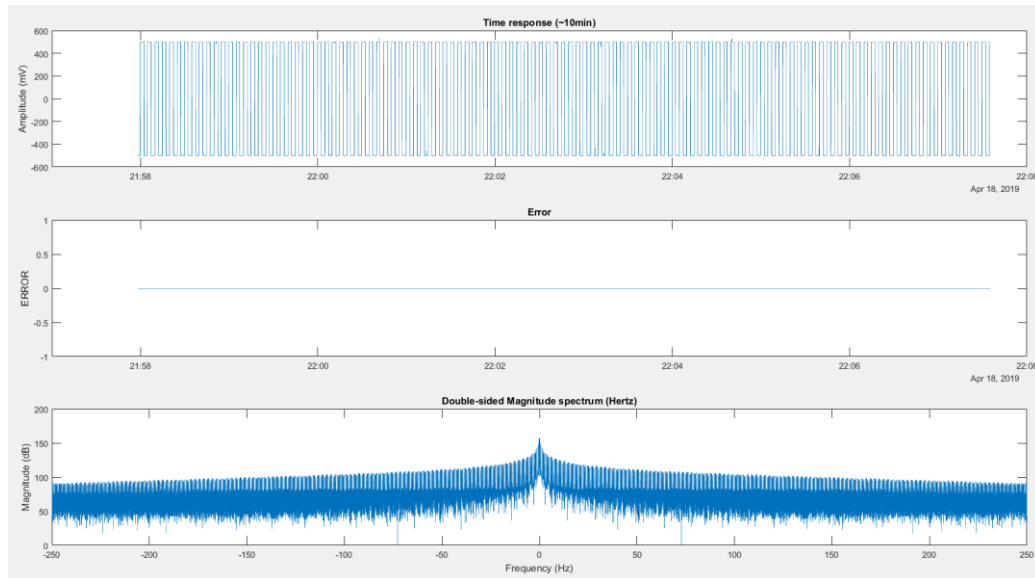
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



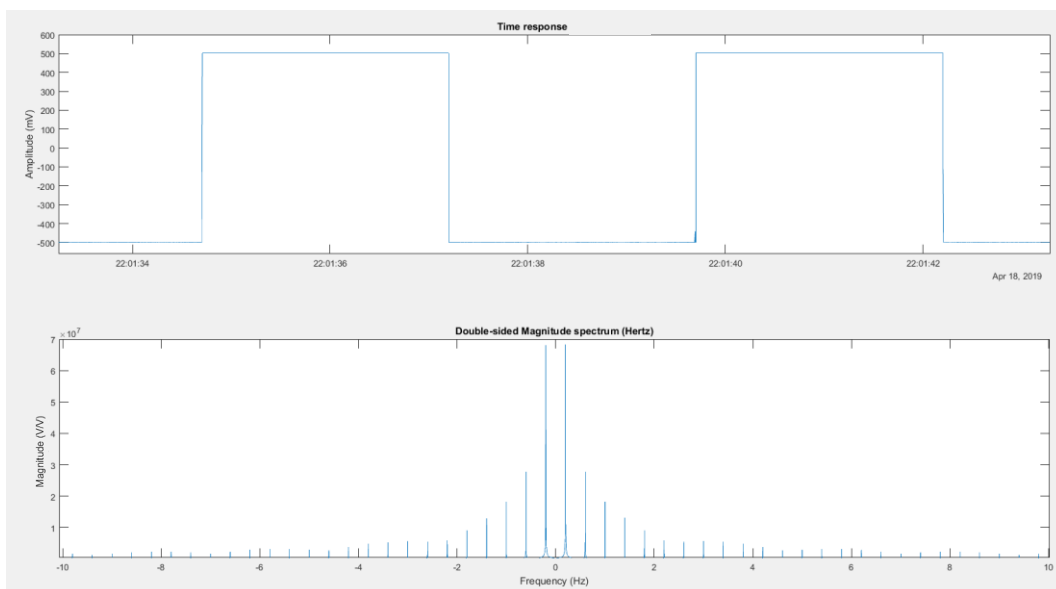
In the time domain, we do observe a period of 0.01 second, therefore a frequency of 100Hz. We observe that the sine wave is no longer really a sine wave. This is due to the fact that a sine at 100Hz is sampled at 500Hz. So we only have 5 samples per period, which is not enough to have a beautiful sinus shape. In the frequency domain, we can see the pulses at 100Hz which represent the sine wave.

Low frequency square wave

A square wave of 0.2Hz and 1Vpp was measured here for a duration of 10 minutes. The results in the frequency and time domain are shown below:



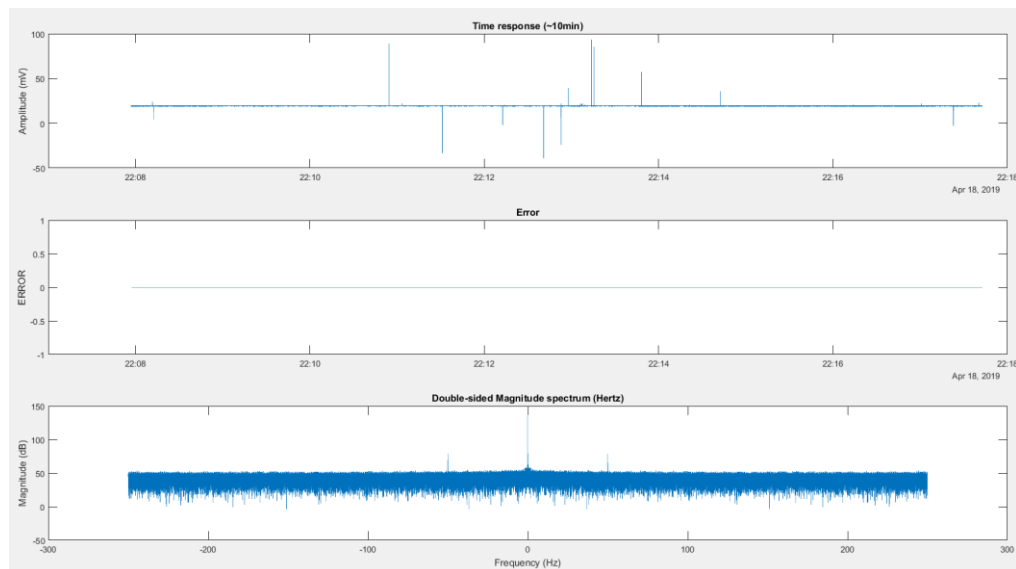
It can be seen that no errors were detected by the device during this period. In the following figure, a zoom was then made to better observe certain details.



In the time domain, we do observe a period of 5 seconds, therefore a frequency of 0.2Hz. In the frequency (here, the result is not in dB !), we can observe that the result is a sinc, which is the fourrier transform of a square wave.

In the air

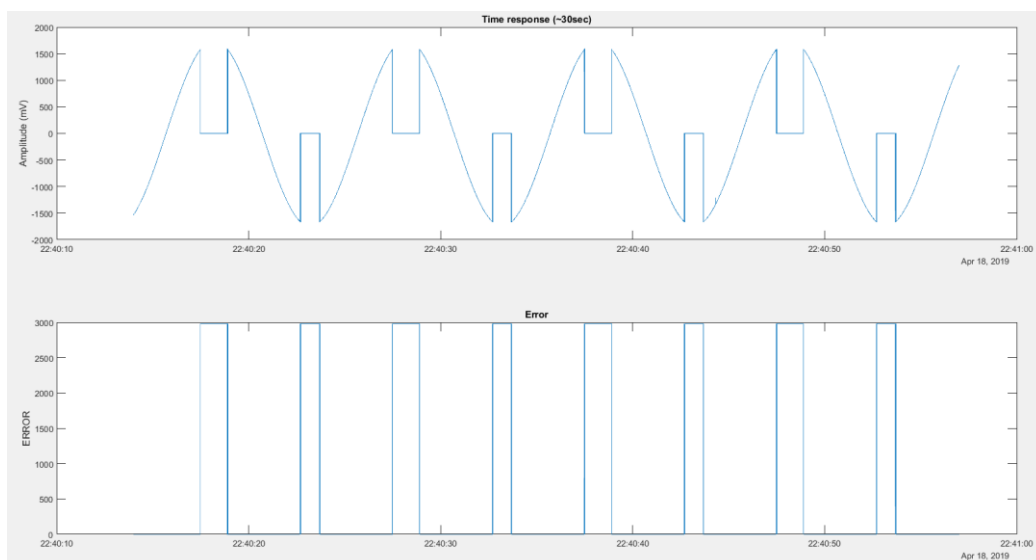
Here, the electrodes were not connected to the function generators and were left in the air to observe what was being measured.



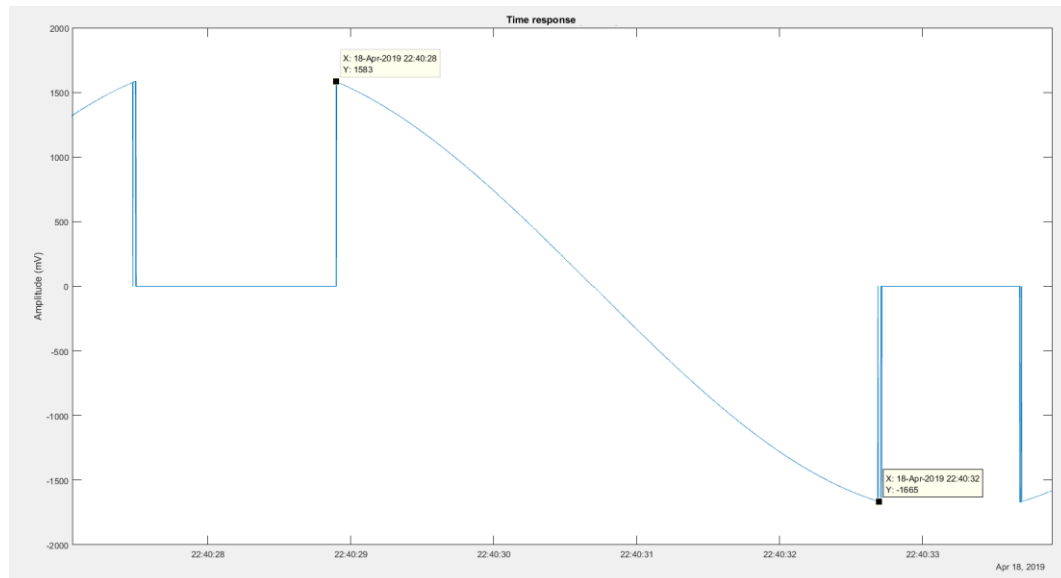
It can be seen that no errors were detected by the device during this period. In the temporal domain, peaks are sometimes observed, certainly due to disturbances on the network and on the cables. Compared to the results in section 4.3, it can be seen that the number of perturbations is lower again. In the frequency domain, we can also observe peaks at 50Hz as well as all these harmonics due to our 50Hz network.

Low frequency sine wave with overvoltage

A sinus wave of 0.1Hz and 3.5Vpp was measured here for a duration of 30 seconds. An overvoltage will therefore be caused each time we exceed $\pm 1.65\text{V}$. The results in the time domain are shown below:



It can be seen that an error is detected each time the limit voltage is exceeded by $\pm 1.65\text{V}$. This has the effect of resulting in a null measurement. A zoom was then performed to see the exact value of this overvoltage :



We then notice that the value of the overvoltage is 1.583V and the undervoltage is -1.665V, which does not exactly correspond to the 1.65V specified in the datasheet. This is certainly due to the AD7770's V_{ss} and V_{dd} power supplies, which are not very accurate.

5 Conclusion

After a few tests, the results are as expected. The various tests confirm that the multi-channel device is working properly.

However, details seem to disrupt the results. First, the range of $\pm 1.65\text{V}$ does not seem quite correct and is asymmetric. This could certainly be improved by using more accurate linear regulators for V_{dd} and V_{ss} .

Secondly and more problematically, several peaks of amplitudes of about 50mV seem to appear in random ways. After the various tests, this may be due to several reasons: the power supply is not completely clean and disruptions occur from time to time. Cables and their contact with the outside (proximity of computer cable, computer power supply, etc. which were relatively close during the test) can also be a source of interference. A long cable length also seems to add slight disturbances. From section 4.5, it does not appear that the cables installed on the plant interfere with each other.

It should be noted that tests for sine waves of 0.01Hz, 1Hz, 100Hz to 50mVpp (with short cables, like in 4.2) were also performed to compare with the use of small signals. These are very similar to those obtained in the previous sections and data are available in the appendices. However, the disturbance peaks were more visible because of the low amplitude.

6 Appendices

- **DHCP_Capture folder:** Folder containing the different captures made for the tests in section 4.1
- **Measures folder:** Folder containing the different data acquired for the tests in section 4.2 and 4.3, 4.4 and 4.5
- **Loading_FFT&T_Maxime.m :** Matlab script for loading data and plotting the measurement in the frequency and time domain